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CE205
Databases and Information Retrieval

Ranked Evaluation and NoSQL (MongoDB)

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Lecturer

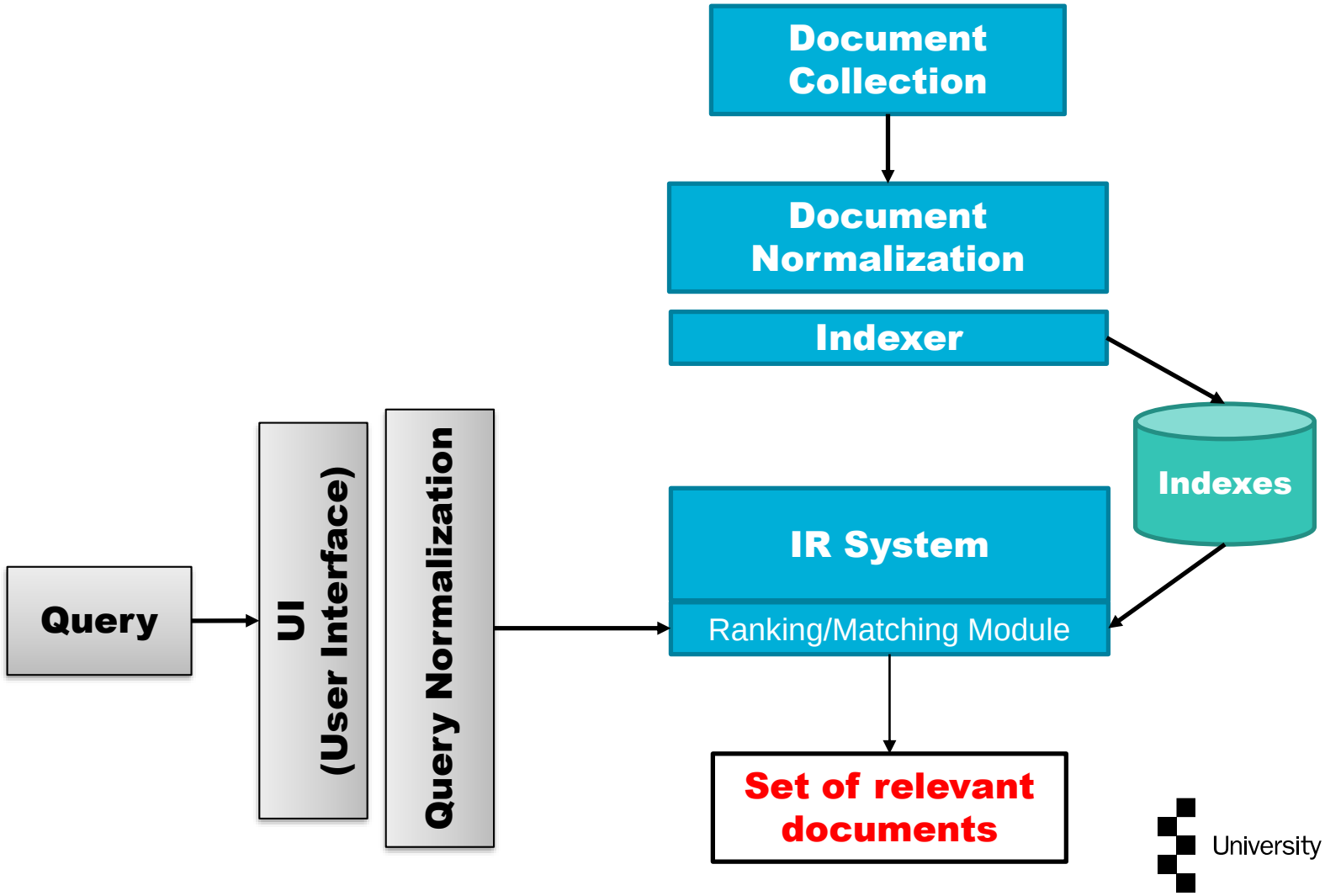
School of Computer Science and Electronic Engineering

- *[Note: Some slides in this set were adapted from an IR course taught by Rada Mihalcea at the University of Michigan and Ray Mooney at UT Austin (who in turn adapted them from Joydeep Ghosh), and from an IR course taught by Chris Manning at Stanford)]*

Learning Objectives

- Ranked Precision and Recall
- Precision at rank p
- Average Precision
- Mean Average Precisions
- Using the Python Database objects to access the MongoDB database
- Implementing the Python MongoDB driver in Python applications
- Connecting to the MongoDB database in Python applications
- Using methods to find and retrieve documents in Python applications
- Sorting documents in a cursor before retrieving them in Python applications

Where were we → Where we are now



Retrieval Effectiveness and Effectiveness evaluation

- Retrieval effectiveness is a measure of **how well the documents retrieved** by a system meet users' needs.

- The **process of determining the retrieval effectiveness** for a given query is referred to as effectiveness evaluation.

General Approach to Effectiveness Evaluation

The general approach to effectiveness evaluation is comprised of the following steps:

(a) User has an information need and expresses this need as a query

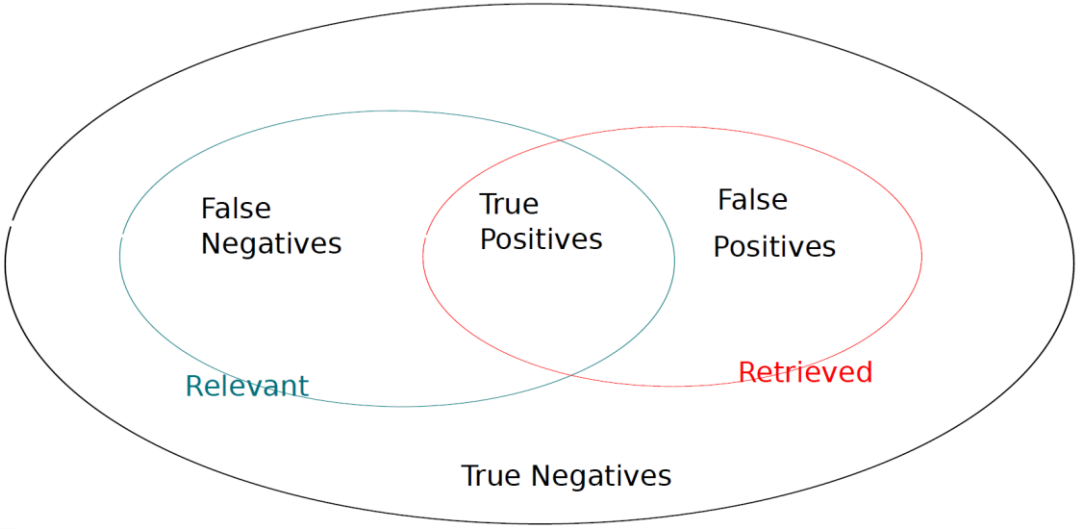
(b) An IR system runs the query against a document corpus, and returns a ranked list of documents

(c) Effectiveness is quantified in terms of how well the ranked list fulfills user's information need

More on Unranked Evaluation

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Evaluation of IR System



		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP (True Positive)**	**FP (False Positive)**
	Not Relevant	**FN (False Negative)**	**TN (True negative)**

Evaluation of IR System: Precision

$$recall = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$recall = \frac{\text{number of relevant documents which are retreived}}{\text{number of relevant documents}}$$

		TRUTH	
		Relevant	Not Relevant
What system think	Retrieved	TP (True Positive)	FP (False Positive)
	Not Relevant	FN (False Negative)	TN (True negative)

Evaluation of IR System: Precision

$$precision = \frac{TP}{TP + FP}$$

$$precision = \frac{\text{number of relevant documents which are retreived}}{\text{number of retreived documents}}$$

		TRUTH	
		Relevant	Not Relevant
What system think	Retrieved	TP (True Positive)	FP (False Positive)
	Not Relevant	FN (False Negative)	TN (True negative)

Exercise 1

Calculate **Precision, Recall, and F1.**

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 20 (True Positive)**	**FP = 40 (False Positive)**
	Not Relevant	**FN = 60 (False Negative)**	**TN = 1,000,000 (True negative)**

Precision and Recall Trade-off

- You can increase recall by returning more docs.
- Recall is a non-decreasing function of the number of docs retrieved.
- A system that returns all docs has 100% recall!
- The converse is also true (usually): It's easy to get high precision for very low recall.

Exercise

An IR system returns 8 relevant documents, and 10 nonrelevant documents. There are a total of 20 relevant documents in the collection. What is the precision of the system on this search, and what is its recall?

F Measure (1)

- F measure is the weighted harmonic mean of precision and recall:

$$F = \frac{1}{\alpha\frac{1}{P} + (1-\alpha)\frac{1}{R}} = \frac{(\beta^2+1)PR}{\beta^2 P + R} \quad \text{where} \quad \beta^2 = \frac{1-\alpha}{\alpha}$$

$$\text{where } \alpha \in [0,1] \text{ and thus } \beta^2 \in [0,\infty]$$

- The default balanced F measure equally weights precision and recall, which means making α = 1/2 or β = 1.
- It is commonly written as F1, which is short for $F_{\beta=1}$

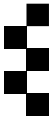
F Measure (2)

- When using β = 1, the formula on the right simplifies to:

$$F_{\beta=1} = \frac{2PR}{P+R}$$

- However, using an even weighting is not the only choice.
  - Values of β < 1 emphasize precision
  - Values of β > 1 emphasize recall
- For example,
- A value of β = 3 or β = 5 might be used if recall is to be emphasized.

Ranked Evaluation



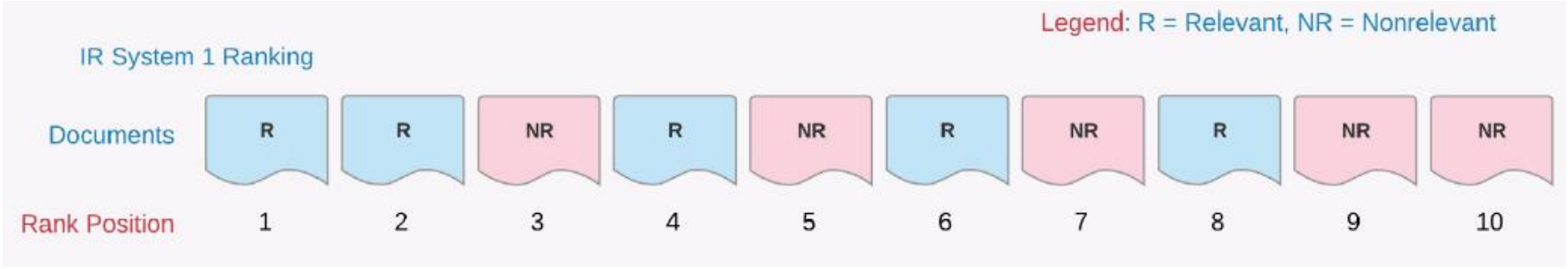
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Ranked Retrieval IR System

- The **definition of precision and recall** assumes that the document collection is partitioned into **two sets** with respect to a query
    - the retrieved set and
    - the rest of the documents (not retrieved set)
- However, **most IR models produce ranked results**.
- **Rank statistics** give some indication of how quickly user will find relevant documents from ranked list

Evaluation for Ranked Retrieval IR System

- The rank of the leftmost document is 1, and the rank numbers increase as we go right.
- Higher the rank number, the lower is the document relevancy.
- However, some nonrelevant documents are placed ahead of relevant documents in the rank ordering.
- In this collection, there are **5 relevant documents** to the query.



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Evaluation for Ranked Retrieval IR System

- The rank of the leftmost document is 1, and the rank numbers increase as we go right.
- Higher the rank number, the lower is the document relevancy.
- However, some nonrelevant documents are placed ahead of relevant documents in the rank ordering.
- In this collection, there are **5 relevant documents** to the query.

Legend: R = Relevant, NR = Nonrelevant

IR System 1 Ranking

Documents	R	R	NR	R	NR	R	NR	R	NR	NR
Rank Position	1	2	3	4	5	6	7	8	9	10
Precision	1.00	1.00	0.67	0.75	0.60	0.67	0.57	0.63	0.56	0.50
Recall	0.20	0.40	0.40	0.60	0.60	0.80	0.80	1.00	1.00	1.00

Precision and Recall at Each Rank Position

For ranked retrieval, we compute precision and recall at each rank position.

- **At rank position one**
  - Precision is 1.00 (since one out of one document(s) is relevant);
  - Recall is 0.20 (since one out of the five relevant documents is retrieved).
- **Likewise, at rank position six**
  - Precision is 0.67 (out of six documents, four are relevant)
  - Recall is 0.80 (four out of the five relevant documents are retrieved).

Precision and Recall at a Rank Position *P* (1)

- Calculation of precision and recall at every rank position is **unsuitable**
  - if the system retrieves a large number of relevant documents for a query

  OR

  - the relevant documents are interspersed with nonrelevant ones throughout the rank ordering
- One solution to this problem is to compute precision and recall only at a small number of predefined rank positions.

Precision and Recall at a Rank Position *P* (2)

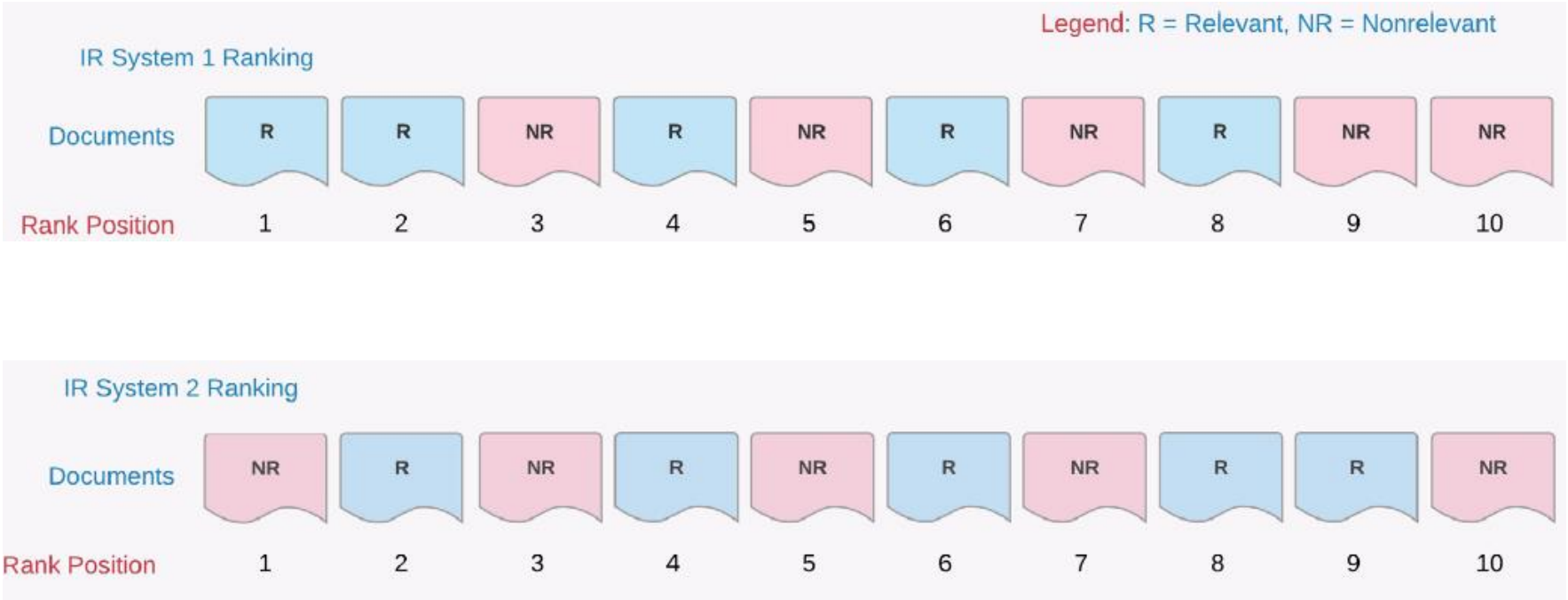
Under this approach, comparing effectiveness of two IR systems requires only **precision**.

- If the **precision** of an **IR System 1** at rank position ***p*** is greater than that of the IR System 2 at the same position, then the **recall** of the **IR System 1** will also be greater than that of the IR System 2.

For example, at the rank position 4

- **IR System 1**, precision and recall are 0.75 and 0.60
- **IR System 2**, precision and recall are 0.50 and 0.40.

Two Ranked Retrieval IR Systems



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Two Ranked Retrieval IR Systems

Legend: R = Relevant, NR = Nonrelevant

IR System 1 Ranking

Documents	R	R	NR	R	NR	R	NR	R	NR	NR
Rank Position	1	2	3	4	5	6	7	8	9	10
Precision	1.00	1.00	0.67	0.75	0.60	0.67	0.57	0.63	0.56	0.50
Recall	0.20	0.40	0.40	0.60	0.60	0.80	0.80	1.00	1.00	1.00

IR System 2 Ranking

Documents	NR	R	NR	R	NR	R	NR	R	R	NR
Rank Position	1	2	3	4	5	6	7	8	9	10
Precision	0.00	0.50	0.33	0.50	0.40	0.50	0.43	0.50	0.56	0.50
Recall	0.00	0.20	0.20	0.40	0.40	0.60	0.60	0.80	1.00	1.00

Precision and Recall at a Rank Position P (3)

- Though many values are possible for the rank position *p*.
- Precision @10 and precision @20 are most commonly used.
- This measure does not differentiate rank orderings for two systems for positions 1 through *p* as seen in both systems
    - precision @10 is the same for both systems

Exercise

Calculate precision and recall at rank position 3, 5, 8 and 9.

Assume that total number of relevant documents in the collection are 5.

Rank	Doc
1	d_{12}
2	d_{123}
3	d_4
4	d_{57}
5	d_{157}
6	d_{222}
7	d_{24}
8	d_{26}
9	d_{77}
10	d_{90}

Blue documents are relevant.

Average Precision (1)

- Average precision, is widely used in practice for measuring the effectiveness of a ranking.
- It involves **averaging the precision** values that correspond to rank positions where a relevant document is retrieved.
- These are the rank positions where recall values increase.

Average Precision (2)

For example

For the IR System 1, such rank positions (where relevant documents are retrieved) are 1, 2, 4, 6, and 8.

The average precision for IR System 1 ranking is

(1.00 + 1.00 + 0.75 + 0.67 + 0.63) / 5 = 0.81

Legend: R = Relevant, NR = Nonrelevant

IR System 1 Ranking

Documents	R	R	NR	R	NR	R	NR	R	NR	NR
Rank Position	1	2	3	4	5	6	7	8	9	10
Precision	1.00	1.00	0.67	0.75	0.60	0.67	0.57	0.63	0.56	0.50
Recall	0.20	0.40	0.40	0.60	0.60	0.80	0.80	1.00	1.00	1.00

Average Precision (3)

- The advantage of the average precision measure is that it provides just one number.

- Though this number is profoundly **influenced by highly ranked documents**, this measure is appropriate for situations that value retrieving as many relevant documents as possible and consider the top-ranked documents as the most important/relevant documents (e.g., web search).

Exercise

Calculate the average precision for the given ranked retrieval IR system.

Legend: R = Relevant, NR = Nonrelevant

IR System 2 Ranking

Documents	NR	R	NR	R	NR	R	NR	R	R	NR
Rank Position	1	2	3	4	5	6	7	8	9	10
Precision	0.00	0.50	0.33	0.50	0.40	0.50	0.43	0.50	0.56	0.50
Recall	0.00	0.20	0.20	0.40	0.40	0.60	0.60	0.80	1.00	1.00

Exercise

Calculate the average precision for the given ranked retrieval IR system.

Rank	
1	X
2	
3	X
4	
5	
6	X
7	
8	
9	
10	X
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	X

X represents the relevant documents.

Mean Average Precision (1)

- The measure that **extends the average precision for multiple queries** for a given IR system is mean average precision, which is the average of the average precisions.

- Again, consider both IR systems and assume that the two rankings shown are for the same system but for different queries.

Mean Average Precision(2)

- We have already calculated average precision for the first ranking as 0.81.

- The average precision for the second ranking is:

  (0.50 + 0.50 + 0.50 + 0.50 + 0.56) / 5 =0.512

- So, the mean average precision is

  (0.81 + 0.512) / 2 = 0.661

Note how the average precision is heavily influenced by highly ranked documents by observing the two rankings.

Exercise

Calculate mean average precision using the given ranking of an IR system for wo different queries.

Query 1

Rank	
1	X
2	
3	X
4	
5	
6	X
7	
8	
9	
10	X
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	X

Query 2

Rank	
1	X
2	
3	X
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	X

Precision-Recall Graph

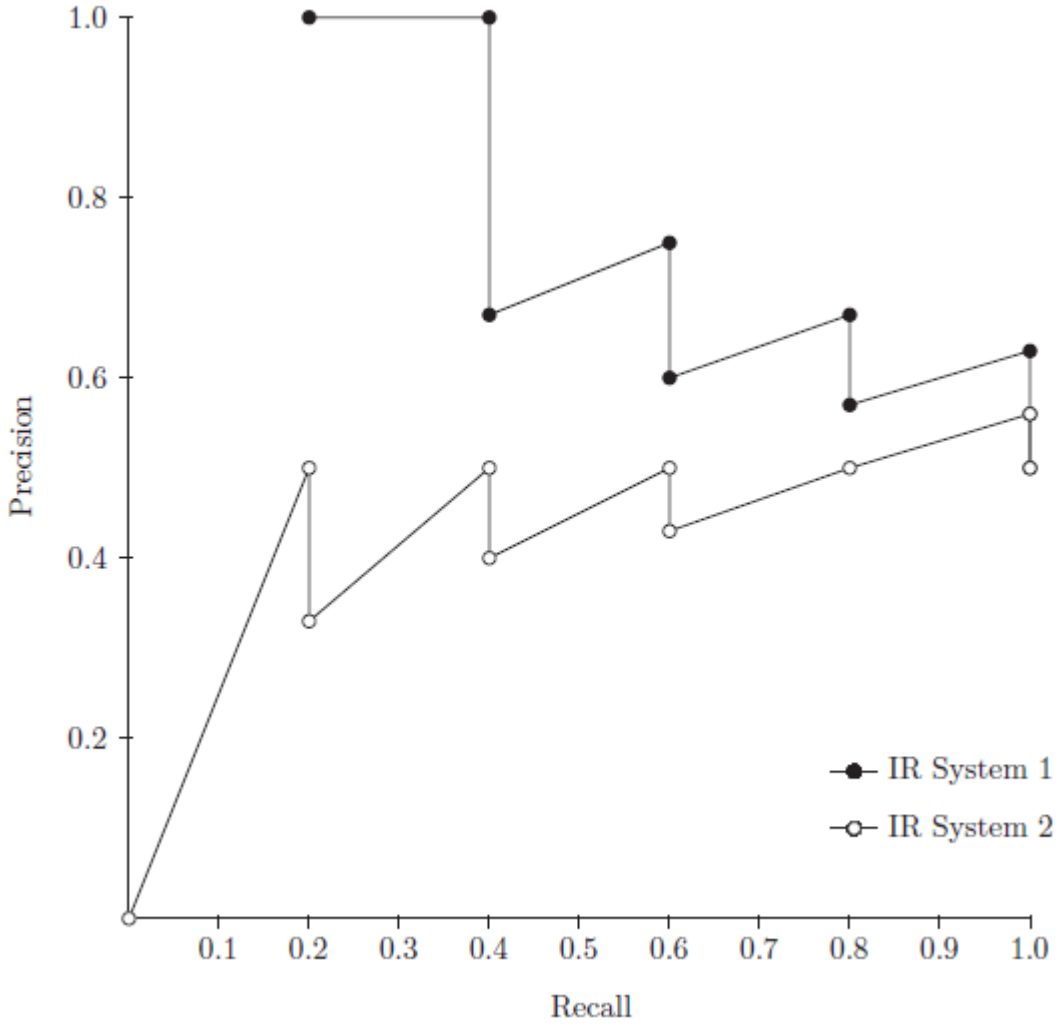
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Precision-Recall Graph

- Though the mean average precision provides effectiveness of a system over several queries using just one number, much information is lost in the process.

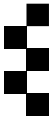
- This is where precision-recall graphs come into play.

Precision-Recall graphs for the Two Rankings



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Kappa Statistic



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Interrater Reliability

- Physician A and Physician B examine 100 patients for swollen knees. Their decisions are summarised in the following table.

		Physician A	
		Yes	**No**
Physician B	**Yes**	**25**	**15**
	No	**5**	**55**

How to test interrater reliability ?

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Kappa Statistic

- A common measure to quantify agreement between various raters/judges is the kappa statistic, which is defined as:

$$\text{kappa} = \frac{p(A) - p(E)}{1 - p(E)}$$

where *p(A)* is the proportion of times the judges agreed, and *p(E)* is the proportion of times they are expected to agree by chance.

Kappa Score Interpretation

Kappa	Agreement
<0	Less than chance agreement
0.01-0.20	Slight agreement
0.21-0.40	Fair agreement
0.41-0.60	Moderate agreement
0.61-0.80	Substantial agreement
0.81-0.99	Almost perfect agreement

Calculate Kappa Statistic

$$\text{kappa} = \frac{p(A) - p(E)}{1 - p(E)}$$

- ***P(A)*** = P(both agreed on Yes) **+** P(both agreed on No)

- ***P(E)*** = Joint Probability of individual raters for Yes
  +
  Joint Probability of individual raters for No

Physician A

Physician B	**Yes**	**No**
Yes	**25**	**15**
No	**5**	**55**

Total Patients= 100

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Calculate Kappa Statistic (2)

$$\text{kappa} = \frac{p(A) - p(E)}{1 - p(E)}$$

- ***P(A)*** = P(both agreed on Yes) **+** P(both agreed on No)

$$P(A) = \frac{25}{100} + \frac{55}{100} = 0.25 + 0.55 = 0.80$$

- ***P(E)*** = Joint Probability of individual raters for Yes

+

Joint Probability of individual raters for No

$$P(E) = \left(\frac{30}{100} \times \frac{40}{100}\right) + \left(\frac{70}{100} \times \frac{60}{100}\right) = (0.12 + 0.42) = 0.54$$

- $kappa = \frac{0.80 - 0.54}{1 - 0.54} = \frac{0.26}{0.46} = 0.56$

Moderate agreement between physicians

Pooling

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Eliciting Relevance Judgments Using Pooling

- This is a manually intensive task, often tedious and error-prone.
- For large collections, manual elicitation of relevance judgments is impractical.
- Pooling is a technique which is used to automate this task.
- For a given query, top *k* items from the rank orderings produced by various IR systems are merged into a pool, duplicates are eliminated, and the items are presented in a random order to experts who perform relevance judgments.

Kappa Statistic in IR System

- Pooling leads to another issue—interrater reliability.
- Relevance judgments for a query provided by multiple experts tend to vary.
- A common measure to quantify agreement between various raters/judges is the kappa statistic, which is defined as:

$$\text{kappa} = \frac{p(A) - p(E)}{1 - p(E)}$$

where *p(A)* is the proportion of times the judges agreed, and *p(E)* is the proportion of times they are expected to agree by chance.

Exercise

Calculate Kappa Statistic

		Judge 2 Relevance		
		Yes	No	Total
Judge 1 Relevance	Yes	300	20	320
	No	10	70	80
	Total	310	90	400

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Suggested Readings

- **Information Retrieval: Concepts, Models and Systems** *in Computational Analysis and Understanding of Natural Languages: Principles, Methods and Applications (Chapter by Venkat N. Gudivada; Dhana L. Rao; Amogh R. Gudivada). Elsevier.*
  - **Section 12: Evaluating IR Systems**

- **Introduction to Information Retrieval** *(Manning, C. D., Raghavan, P., & Schütze, H. (2008). Cambridge University Press.* ***Available online at https://nlp.stanford.edu/IR-book/***
  - **8.3 Evaluation of unranked retrieval sets**
  - **8.4 Evaluation of ranked retrieval results**